

Number Sequences

Worksheet 2 (Non-Calculator)

1. S15/qp11/q26

A pattern of numbers is given below.

$$\text{Row 1} \quad \frac{1}{1 \times 2} = \frac{1}{1} - \frac{1}{2}$$

$$\text{Row 2} \quad \frac{1}{2 \times 3} = \frac{1}{2} - \frac{1}{3}$$

$$\text{Row 3} \quad \frac{1}{3 \times 4} = \frac{1}{3} - \frac{1}{4}$$

$$\text{Row 4} \quad \frac{1}{4 \times 5} = \frac{1}{4} - \frac{1}{5}$$

- a. Write down Row 10. [1]

- b. Adding the first two rows gives the result $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} = \frac{1}{1} - \frac{1}{3} = \frac{2}{3}$

$$\text{Adding the first three rows gives the result } \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} = \frac{1}{1} - \frac{1}{4} = \frac{3}{4}$$

- i. Write down the result of adding the first four rows. [1]

- ii. Use the pattern to write down the value of,

- a. the number of rows that add up to $\frac{109}{110}$, [1]

- b. an expression, in terms of n , for the result of adding the first n rows. [1]

2. S15/qp12/q26

- a. The first four terms of a sequence, S , are 89, 83, 77, 71.

- i. Find an expression for S_n , the n th term of this sequence. [2]

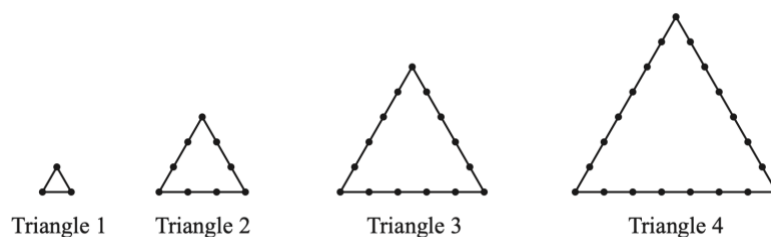
- ii. Find the smallest value of n for which $S_n < 0$. [1]

- b. The n th term of a different sequence, T , is given by $T_n = n^2 - 4n$.

- i. Find and simplify an expression for $T_{n+1} - T_n$. [2]

- ii. The difference between T_{p+1} and T_p is 75. Find the value of p . [1]

3. W16/qp11/q23



The diagrams show a sequence of triangles made up of identical sticks. Each triangle has two more sticks on each edge than its previous triangle. The table shows information relating to this sequence.

Triangle number	1	2	3	4		n
Number of sticks on each side	1	3	5			x
Number of sticks in the triangle	3	9	15			y

- Complete the column for triangle 4. [1]
- Find an expression, in terms of n , for x . [1]
- Find an expression, in terms of n , for y . [1]
- The total number of sticks in the first triangle = 3

The total number of sticks in the first two triangles = 12

The total number of sticks in the first three triangles = 27

- Write down the total number of sticks in the first four triangles. [1]
- Find an expression, in terms of n , for the total number of sticks in the first n triangles. [2]

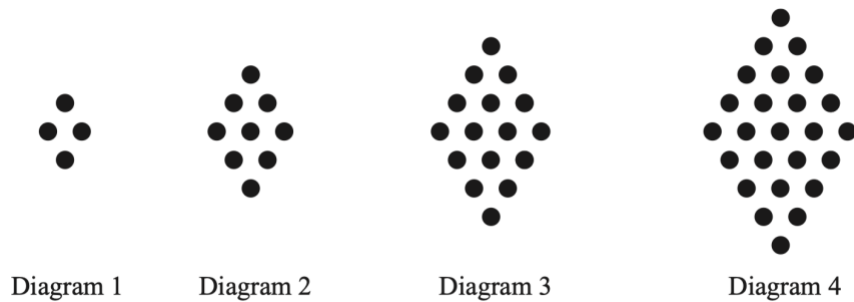
4. S16/qp12/q25

- The n th term of a sequence is given by $n^2 - 5n$.
 - Find the 2nd term in the sequence. [1]
 - The p^{th} term in the sequence is 150. Find the value of p . [2]
- The n th term of another sequence is given by $3n^2 - kn$.

The 5th term in this sequence is 55. Find the value of k . [2]

5. W17/qp11/q25

Mary makes pendants, of the same design, from small beads. The sequence of diagrams shows the pendants she makes.



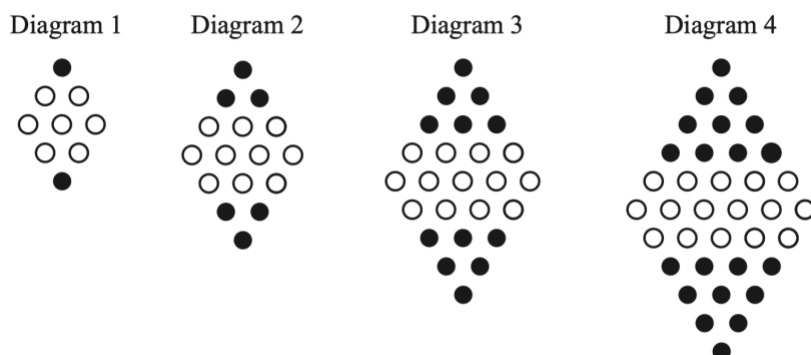
- a. Complete the table. [1]

Diagram number	1	2	3	4	5
Number of rows	3	5	7	9	
Number of beads	4	9	16	25	

- b. Find an expression, in terms of N , for
- the number of rows in Diagram N , [1]
 - the number of beads in Diagram N . [1]
- c. Julia asks Mary to make her a pendant with 25 rows. How many beads are used to make this pendant? [2]

6. W17/qp12/q26

The sequence of diagrams shows patterns made from some black beads and some white beads. Each diagram has two rows more than the previous diagram.



a. Complete the table for Diagram 5.

[1]

Diagram number	1	2	3	4	5
Total number of beads	9	16	25	36	
Number of white beads	7	10	13	16	
Number of black beads	2	6	12	20	

b. Write down an expression, in terms of n , for

i. the number of white beads in Diagram n ,

[1]

ii. the total number of beads in Diagram n .

[1]

c. Find an expression, in terms of n , for the number of black beads in Diagram n . Give your answer in its simplest form.

[2]

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Answer Key

1. S15/qp11/q26

26 (a)	$\frac{1}{10 \times 11} = \frac{1}{10} - \frac{1}{11}$	1	
(b) (i)	$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} = \frac{1}{1} - \frac{1}{5} = \frac{4}{5}$	1	
(ii) (a)	$\frac{19}{20}$	1	
(b)	109	1	
(c)	$\frac{n}{n+1}$ oe	1	

2. S15/qp12/q26

26 (a) (i)	$95 - 6n$ oe isw	2	B1 for $-6n$ seen
(ii)	16 cao	1	
(b) (i)	$2n - 3$	2	M1 for $(n+1)^2 - 4(n+1)$ seen
(ii)	39 cao	1	

3. W16/qp11/q23

23 (a)	7, 21	1	
(b)	$2n - 1$ oe	1	
(c)	FT $3 \times$ their (b) provided this is a function of n ; or $6n - 3$ oe	1 ✓	
(d) (i)	48	1	
(ii)	$3n^2$	2 *	M1 for a sensible method, e.g. writing terms as $3 \times 1, 3 \times 4, 3 \times 9, \dots$ or B1 for $An^2 + Bn + C, A \neq 0$ from a valid method.

4. S16/qp12/q25

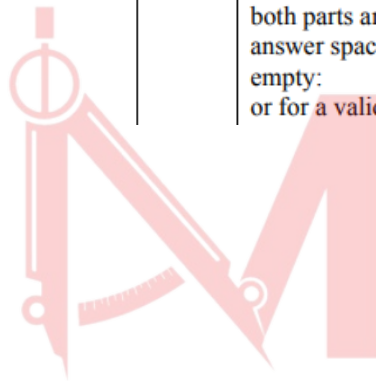
25 (a) (i)	-6	1	
(ii)	15	2*	C1 for $15^2 - 5 \times 15$ or for 15, -10 OR M1 for $(p+10)(p-15) [= 0]$
(b)	4	2*	B1 for $3 \times 5^2 - 5k = 55$ oe

5. W17/qp11/q25

25(a)	11, 36	1	
25(b)(i)	$2N+1$	1	
25(b)(ii)	$(N+1)^2$ oe	1	
25(c)	169	2	B1 for <i>their (b)(i)</i> = 25; or for $N = 12$

6. W17/qp12/q26

26(a)	49, 19, 30	1	
26(b)(i)	$3n+4$ oe and isw	1	
26(b)(ii)	$(n+2)^2$ oe	1	
26(c)	n^2+n ; or $n(n+1)$	2	M1 for attempt at <i>their(bii)</i> – <i>their(bi)</i> , provided both parts are different expressions in n , and the answer space also contains an expression in n , or is empty: or for a valid method.



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